

CREATING DISPLAYS FOR NUMERICAL DATA



LESSON INTRODUCTION

MA.7.DP.1.5 Given a real-world numerical or categorical data set, choose and create an appropriate graphical representation.

LEARNING TARGETS

- I can create an appropriate graphical representation for a numerical data set.

BENCHMARK COHERENCE

BUILDING ON

MA.6.DP.1.5 Create box plots and histograms to represent sets of numerical data within real-world contexts.

WORKING TOWARD

MA.912.DP.1.1 Given a set of data, select an appropriate method to represent the data, depending on whether it is numerical or categorical data and on whether it is univariate or bivariate.

ESSENTIAL QUESTIONS

- What are the different types of graphical representations of numerical data?
- How can you choose the best graphical representation of a data set?
- Why are some graphical representations better for certain data sets than others?

LESSON NARRATIVE

In this lesson, students apply what they learned in Lesson 7 to choose and create appropriate graphical representations for given data sets. Students begin by reading about a data set and choosing between a histogram and line plot to best represent the data. Students then explain their choice and why it is best suited for the data presented. The content progresses as students then create their own graphical displays from a data set and evaluate their responses with a partner, comparing and determining whose representation best reflects the data set.

COMPONENT SUMMARY

13.8.1 WARM-UP (~5 MIN.)

Find missing values in a data set

13.8.3 TRY IT (10-15 MIN.)

Create graphical representations of data sets with a partner

13.8.5 WRAP-UP (~5 MIN.)

Students reflect on the different type of displays that can be used to represent numerical data and identify what their favorite display is

13.8.2 COLLABORATION (10 MIN.)

Create graphical representations of data sets

13.8.4 YOUR TURN! (10-15 MIN.)

Create graphical representations of data sets and compare independently

RESOURCES

GLOSSARY

Key Terms: N/A

Additional Terms: N/A

MATERIALS

- N/A

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may struggle to choose which graphical representation is best for each data set.

THINGS TO CONSIDER

- Consider allowing students to use the reference sheet created in the previous lesson as they work through each section.

LITERACY AND LANGUAGE ROUTINES

Think-Pair-Share

13.8.3 Try It provides an opportunity to engage students in a Think-Pair-Share routine. Students have the opportunity to verbalize their explanations internally and then with a partner before having the opportunity to share with the class. Sharing in a low-stakes capacity allows students to build confidence in their response before formalizing it in writing.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.4.1 Discussion and Reflection

This lesson contains several activities where students will work collaboratively with a partner to determine the best graphical displays for several different sets of data. Through these activities, students will engage in collaborative discussion and reflect on their findings when comparing their choices and determining the best displays.

DIFFERENTIATE INSTRUCTION

13.8.2 Collaboration has students working together in partner pairs. Consider pairing students based on the data from the previous lesson. This can help struggling students learn from their peers, while strengthening the other students' understanding of the concept.

Each part of this lesson requires students to create graphical displays of the data presented. This may be difficult for some students. Students should be given graph paper and access to a computerized way to create the graphical representations, as needed. These alternative means of representation provide additional entry points and ways for students to represent their understanding.

13.8.1 WARM-UP: MYSTERY BOXES

TEACHER MOVES

Use student understanding to inform your instruction and to determine where review of skills is needed. Make a list of misconceptions and determine which should be addressed whole group or small group.



13.8.1 Warm-Up: Mystery Boxes

- The data set below contains 14 numbers, listed in order from least to greatest. Six of the values have been removed. Your job is to use the clues given to determine the missing values. Write your answers in the boxes.

- The mean is 26.
- The median is 22.
- The interquartile range is 20.
- The mode is 18.
- The range is 60.

1	3	4	12	15	18	18	26	29	30	32	57	58	61
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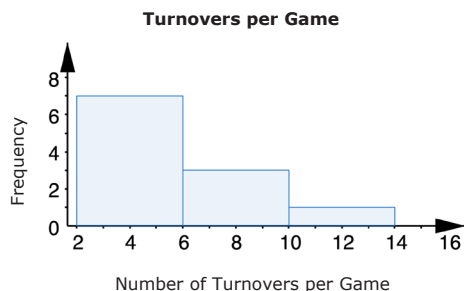


13.8.2 Collaboration: Who Made it Best?

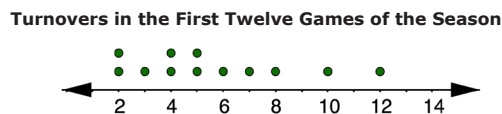
1. Dante and Lorraine are both managers of the seventh-grade basketball team. They are trying to decide which graphical representation to use to show the number of turnovers the team has had per game in the first 12 games of the season, shown in the data set below.

{2, 2, 2, 3, 4, 4, 4, 5, 5, 6, 7, 8, 10, 12}

Lorraine made a histogram, as shown.



Dante made a line plot, as shown.



- Which of the two graphical representations shown is best to display the data?
Answers will vary. Dante's line plot provides more specific information.
- Explain your reasoning.
Answers will vary. Dante's line plot shows each game's number of turnovers. It is clear to see that in one of the games the team had 12 turnovers.
- Compare your work with your partner and if necessary, resolve any differences.
- Discuss with your partner whether a box plot or stem-and-leaf plot would be useful for displaying this data set. Summarize your discussion.
Answers will vary. A box plot would provide a 5 number summary of the data and would better show the IQR clearly.

13.8.2 COLLABORATION: WHO MADE IT BEST?

TEACHER MOVES

Students may struggle to determine which graphical representation best displays the data. Remind students:

- Histograms are best for continuous data.
- Line plots are good for data sets with small ranges. They show details of the data well, like gaps, each data point, and the shape of the data.
- Box plots are good for comparing one data set to another, and for data with a large range. They show the five-number summary, but you cannot see each data point. While box plots do give a good idea of the shape of the data, you cannot see gaps from a box plot, so if there are gaps in the data, that detail can be lost. Box plots make outliers obvious too.
- Stem-and-leaf plots are quite similar to line plots, but they are less effective with data that has a very small range.

Consider referring students to the Bring It Together from Lesson 7 as a reference when completing this component.

DIFFERENTIATE INSTRUCTION

Consider providing sentence starters for questions b and c. This provides an additional scaffold for ELL students, as well as students who may struggle to express themselves in writing.

ENRICHMENT

Students may struggle to articulate which graphical representation is most useful for displaying the data set. Provide guidance through questioning:

- What type of data are you displaying?
- What is the range of data? Are there any gaps or outliers?
- Which representation gives the most specific information?

13.8.3 TRY IT: CREATING GRAPHICAL REPRESENTATIONS

TEACHER MOVES

Students may be unsure about which display to construct because an outlier is included in the data set. A box plot is appropriate due to the large range of values caused by the outlier.

DIFFERENTIATE INSTRUCTION

Provide an additional auditory entry point for students through partnership. One partner reads through each scenario and question set aloud to provide increased accessibility to the material. This will be especially beneficial for ELLs or struggling readers.

TEACHER MOVES

Students will choose their display based on the number of data values and the scale used for the display. Since the data is rounded to the tenths place, a histogram or box plot should be chosen over a line plot.

QUESTIONING

If students finish early, encourage them to create the additional display they named for question 2c to provide additional practice with the lesson's learning target.



13.8.3 Try It: Creating Graphical Representations

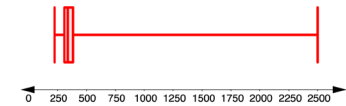
1. Laverne and Rocco planned to travel the greatest distance between any two points in the United States of America — from Log Point, Elliot Key, Florida to Kure Atoll, Hawaii — a distance of 5,859 miles.

They took a ferry from Log Point to Miami, then drove to Los Angeles, California, where they took a plane to Hawaii. They drove anywhere from 5-8 hours a day, and kept track of how many miles they traveled per day on their trip. This information is shown below.

{224, 294, 309, 317, 321, 339, 355, 361, 383, 456, 2500}

- a. Create an appropriate graphical representation for the data.

Answers will vary.

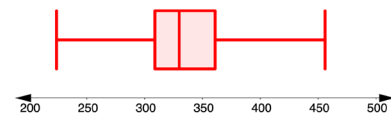


- b. If the value 2500 were removed from the data set, which display would you create to represent the data? Explain your reasoning.

Answers will vary. A box plot would provide the five-number summary of the data. A histogram would display the shape.

- c. Create an appropriate graphical representation for the data without the value 2500.

Answers will vary.



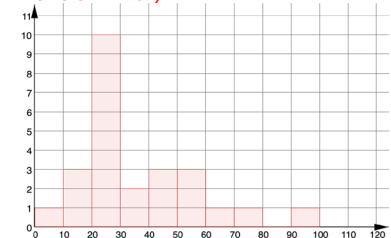
2. An earthworm farmer set up several containers of a certain species of earthworm so he could learn about their lengths. Lengths of earthworms provide information about their ages. The farmer measured the lengths of the earthworms in one container.

The farmer recorded the earthworms' lengths, in millimeters (mm), in the table shown.

Earthworm Lengths in Millimeters (mm)										
6.0	11.2	18.3	19.4	20.2	23.1	23.3	25.2	25.7	26.0	
27.8	27.9	28.4	29.9	32.5	33.2	41.1	42.3	48.5	52.1	
54.1	59.7	60.3	77.6	93.2						

- a. Create an appropriate graphical representation for the data.

Answers will vary.



- b. Explain your reasoning for choosing your display type.

Answers will vary. The histogram was chosen because the intervals allow for decimal variation.

- c. If you were asked to create an additional display, which would you make and why?

Answers will vary. A box plot would also be an appropriate display because it provides a 5 number summary and eliminates the need for a large range of numbers to accommodate the decimals.



13.8.4 Your Turn!

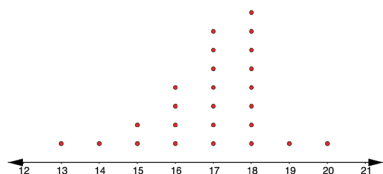
1. The birth weights, in ounces, of golden retriever puppies born at a kennel were recorded for the past month. The data is recorded in the table.

13	14	15	15	16	16	16	16	17	
17	17	17	17	17	17	18	18	18	
18	18	18	18	18	19	20			



- a. Create an appropriate graphical display for the data.

Answers will vary.



- b. Explain your reasoning for choosing that display type.

Answers will vary. The line plot was selected because the range of the data was small. The line plot allows for the shape of the distribution to be determined.

13.8.4 YOUR TURN!

ENRICHMENT

If students are struggling to choose an appropriate graphical display for the data, ask questions to provide additional scaffolding:

- Based on the range of this data, which graphical representation would be a good fit for it?
- Is there any graphical representation that you can tell wouldn't be a good fit for the data?
- Which graphical representation would help us to see the shape of this data?

13.8.4 YOUR TURN! (CONTINUED)

TEACHER MOVES

Students may try to use the year values in the display. This display should only summarize the pass-completion percentages. A histogram or box plot are also appropriate to display the data.

DIFFERENTIATE INSTRUCTION

Consider alternative means for creating the graphical displays. Some students may benefit from the use of graph paper to keep their displays organized. To reduce barriers to learning created by motor demands of this task, it may be helpful to provide students with digital alternatives for creating their displays.

2. Tom Brady's pass-completion percentages from 2009 to 2020 are listed in the table.

Year	2009	2010	2011	2012	2013	2014
%	65.7	65.9	65.6	63	60.5	64.1
Year	2015	2016	2017	2018	2019	2020
%	64.4	67.4	66.3	65.8	60.8	65.7

- a. Create an appropriate graphical display for the data.

Stem	Leaf
60	5, 8
61	
62	
63	0
64	1, 4
65	6, 7, 7, 8, 9
66	3
67	4

Scale: 60|5 = 60.5%

- b. Explain your reasoning for choosing that display type.
Answers will vary. Stem-and-leaf plot was chosen to represent Tom Brady's pass percentage so that each individual year could be represented. Even though a line plot also displays the individual data values, it would have been difficult to construct because of the scale
- c. Compare your graph with your partner. If you created the same type of graph, explain why this graph is the best choice to represent the data. If you created different displays, discuss which graph is a better choice.

**13.8.5 Wrap-Up: Cool Down**

1. What is your favorite type of graphical representation to create when displaying numerical data?
Answers will vary.
2. Write a note to your teacher explaining why this is your favorite type of display to create. Be specific in your explanation. You can use illustrations and examples from the lesson to help in your explanation.
Answers will vary.

13.8.5 WRAP-UP: COOL DOWN**DIFFERENTIATE INSTRUCTION**

Consider alternative ways for students to complete the Cool Down. Some students may benefit from verbally explaining the lesson to a partner or recording their response using an app like Flipgrid.